

A Low-Complexity Return-Link Beam-Hopping Scheduling for NGSO Mega-Constellations With Dynamic Topology and Uneven Traffic

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Abstract—Nongeostationary orbit (NGSO) constellations characterized by seamless coverage and high throughput have been envisioned as a promising solution to 6G, where beam hopping (BH) technique plays an indispensable role to achieve on-demand service. However, owing to the dynamic topology and uneven terrestrial traffic, how to effectively manage multisatellite time-frequency-spatial beam resources to match the heterogeneous demands is challenging for NGSO satellite operators. This article proposes a novel return-link BH scheduling framework for NGSO mega-constellations, where the ground resource management center periodically executes BH scheduling for users in Earth-fixed cells to improve the long-term average service satisfaction (ASS) while reducing the intersatellite handover. Specifically, we decouple the NP-hard scheduling problem into three subproblems, namely, user-satellite association problem, BH pattern design problem and resource block (RB) allocation problem. User-satellite association problem is solved by a low-complexity heuristic algorithm to reduce the handover and avoid intersatellite interference. Furthermore, a graph-based maximum weight method and an iterative scheme based on successive convex approximation are developed to obtain the BH patterns and RB allocations, respectively. Simulation results demonstrate that compared with other benchmarks, such as the best channel association and round-robin BH schemes, the proposed method can reduce the satellite load unevenness by 60% and improve the ASS by 35% while maintaining a low handover overhead.

Index Terms—Beam hopping (BH), intersatellite handover, nongeostationary orbit (NGSO) constellations, resource management, return link.

I. INTRODUCTION

RECENTLY, we have witnessed the rapid development of the nongeostationary orbit (NGSO) mega-constellations due to its advantages, such as low latency, extensive coverage, and ultrawideband capabilities [1], [2]. Specifically, some low-Earth orbit (LEO) constellations represented by Starlink,

Oneweb, and Kuiper are undergoing accelerated deployment with the aim of achieving seamless global coverage, providing broadband transmission services and addressing the massive connectivity of Internet of Things (IoT) [3]. As a result, some organizations, such as 3rd Generation Partnership Project (3GPP) and European Telecommunications Standards Institute (ETSI) have conducted standardization for nonterrestrial networks (NTNs) and the NGSO satellite systems are envisioned as one of the promising solution to beyond fifth generation (B5G) or sixth generation (6G) [4], [5]. However, owing to the geographically nonuniform distribution of terrestrial users, the traffic demands within satellite coverage areas are time-varying, showing significant challenges for satellite resource management. Moreover, due to the high mobility of NGSO satellites, traditional fixed-point multibeam satellite systems suffer the severe interbeam handover. Therefore, flexible beam coverage becomes a key challenge for NGSO systems.

Beam hopping (BH) technique has been considered as a crucial solution to address the aforementioned challenges. Leveraging the rapid beamforming capability of phased-array antennas, BH satellite systems can generate multiple spotbeams with high gain and flexibly change the direction of the beams to match the uneven traffic demands. Besides, the spotbeams can be steered toward users, reducing the user handover compared to conventional fixed antenna. In view of the high flexibility of BH, some standards, such as 5G-NTN and DVB-S2X, have already identified BH systems as a future trend in NTN development [6], [7].

In order to fully exploit the potential of BH, efficient resource management plays an important role, which has attracted many scholars to conduct extensive research. Alegre-Godoy et al. [8] proposed an iterative BH scheme based on maximizing the signal-to-interference-plus-noise ratio (SINR). Besides, Zhang et al. [9] investigated a BH scheduling method by Cuckoo search to maximize the weights of users. Similarly, Wu et al. [10] proposed a maximal service weight gain BH algorithm to improve Quality of Service (QoS) of users. Additionally, some heuristic algorithms are widely adopted in BH scheduling. Gou et al. [11] investigated a low-complexity heuristic return-link BH method based on maximum-weighted clique (MWC) algorithm for geostationary Earth orbit (GEO) satellite, where the traffic distribution, interbeam interference and channel condition are considered.

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Moreover, an efficient beam capacity utilization problem was formulated in [12] and solved by a genetic algorithm (GA). Meanwhile, Wei et al. [13] proposed a dynamic BH scheduling strategy using enhanced artificial bee colony algorithm to accelerate the calculation convergence. To mitigate the interbeam co-channel interference, some works, which combine precoding with BH, have been proposed in [14], [15], [16], [17], and [18]. Gupta et al. [15] proposed a flexible cluster hopping approach where a few clusters are on-demand illuminated and the precoding is designed to improve the beam capacity. However, these precoding-based methods require accurate channel status. In addition, some literature introduces nonorthogonal multiple access (NOMA) into BH to further increase system capacity [19], [20]. According to [20], the power allocation, carrier assignment, and beam scheduling are jointly optimized to satisfy uneven traffic demands in the NOMA-based satellite systems. With the application of machine learning (ML) in satellite networks, some scholars used ML to empower BH scheduling [21], [22], [23], [24], [25], [26]. Lei et al. [21] proposed an optimization approach combined with ML to achieve a rapid BH scheduling. A cooperative multiagents deep reinforcement learning (MADRL) framework was proposed in [24] to achieve joint BH patterns and bandwidth allocation. Besides, Hu et al. [22] explored a new BH framework for GEO satellite where the GA approach was powered by DRL.

However, above-mentioned BH algorithms are mainly designed for a single satellite, which are not suitable for the NGSO constellations with multiple satellites and high dynamics. Therefore, several BH scheduling methods for NGSO satellite network have been investigated. According to [27], a low-complexity user density-based BH design scheme was presented to minimize the number of beam positions. Lin et al. [28], [29] proposed a multisatellite BH algorithm based on load balancing and interference avoidance to improve the service satisfaction of Earth-fixed cells while considering the GEO interference avoidance. Moreover, Li et al. [30] proposed a power allocation competition algorithm to maximize the weighted capacity-request ratio for cognitive LEO BH satellite-terrestrial networks. Besides, a 5G-based LEO satellite communication system was considered in [31], where BH patterns and satellite transmit power are jointly scheduled. Furthermore, Wang et al. [32] carried out rigorous system-level simulations for new radio (NR) enabled LEO constellations where the round-robin BH scheme and the classical proportional fairness (PF) scheduler were adopted.

Although the above works have studied the BH scheduling for the purpose of improving QoS, reducing the interbeam interference and ensuring user fairness, some specific issues in NGSO networks are ignored. First, due to the high-speed motion of satellites, the topology between satellites and users changes rapidly, causing time-varying intersatellite interference and satellite-ground connection. Thus, the GEO-based BH scheduling methods are not suitable to this dynamic topology scenario. Second, since the visual relationship between satellites and ground users changes frequently, BH scheduling would be updated in time. However, existing BH scheduling algorithms for NGSO constellations only

consider the instant snapshot topology rather than the impact of long-term topological changes, which can easily lead to overmuch intersatellite handover for users and severely reduce the communication capacity. Third, most literature focuses on cell-based beam scheduling where each cell is regarded as the minimum allocation unit, which may be infeasible in reality. Actually, there are multiple users located in one cell. How to associate users with satellites and how to allocate beam resource to users with heterogeneous demands are the inevitable problems for BH scheduling in dynamic satellite network. Finally, despite extensive research on forward-link BH, there is a lack of works addressing the BH scheduling in return link transmission scenario. Traditional return link in satellite communication adopts fixed multibeam receiving antennas, which has the disadvantages of large number of radio frequency (RF) channels, high power consumption, and inflexible pointing. Combining BH with return link can significantly reduce power consumption due to fewer RF chains, and improve the flexibility of beam pointing, especially in the NGSO constellation scenario, where the high dynamic motion of the satellite requires timely adjustment of the beam direction to stare at the user. To the best of the authors' knowledge, there is currently no research on the joint design of BH and multiple access for return-link transmission in NGSO mega-constellation communications.

Motivated by the above discussions, this article proposes a universal return-link BH scheduling architecture for NGSO mega-constellation communications with dynamic topology and uneven traffic, aiming at enhancing user service satisfaction while reducing the intersatellite handover. In order to make the primal BH scheduling problem more tractable, this article decouples it into three subproblems, namely, user-satellite association problem, BH pattern design problem and resource block (RB) allocation problem. Particularly, these subproblems are solved by the ground resource management center per scheduling epoch for practical implementation. The key contributions and innovations of this article are summarized as follows.

- 1) Presentation of a practical return-link BH resource management architecture for NGSO mega-constellations. In this framework, a ground resource management center periodically conducts BH scheduling for users in Earth-fixed cells of a service region, which considers the dynamic network topology, time-varying channel, interbeam interference, nonuniform user distribution, and multiple access of users.
- 2) Formulation of the dynamic multisatellite BH scheduling problem, aiming at maximizing the long-term average actual service satisfaction of users with the consideration of handover overhead. For the purpose of simplifying this NP-hard problem, it is decomposed into three subproblems, including user-satellite association, BH pattern design, and RB allocation.
- 3) Presentation of low-complexity algorithms to solve three subproblem. First, a heuristic user-satellite association strategy is proposed to avoid intersatellite interference, balance intersatellite load and reduce intersatellite handover. Second, a graph-based maximum dynamic weight

method is presented to determine the BH pattern slot-by-slot. Third, we propose an iterative algorithm based on successive convex approximation (SCA) to implement RB and power allocation. Moreover, the complexity analysis shows that our proposed algorithm has the polynomial-time complexity.

- 4) Presentation of extensive simulation results under different user distribution. Actual constellation configurations and system settings are adopted. Numerical experiments validate the superiority of the proposed method compared with different benchmarks in terms of service satisfaction, user fairness, handover overhead, and load balancing.

The remainder of the work is organized as follows. Section II describes the system model and formulates the dynamic return-link BH scheduling problem. Section III decomposes the original problem into subproblems and proposes the low-complexity solutions. In Section IV, simulation results and performance analysis are elaborated under different baselines, followed by conclusions in Section V. The detailed notations and definitions used in this article are summarized in Table I.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As shown in Fig. 1, we specifically consider the return-link transmission (e.g., the satellite IoT scenario) in the multibeam NGSO constellation communication system, where a specific service area, consisting of M Earth-fixed cells, is operated by a ground resource management center (RMC). The set of cells and satellites are denoted by $\mathcal{M} = \{m \mid m = 1, 2, \dots, M\}$ and $\mathcal{N} = \{n \mid n = 1, 2, \dots, N\}$, respectively. Assume there are U user equipments (UEs) located in this area, and denote $\mathcal{U} = \{u \mid u = 1, 2, \dots, U\}$ as the set of UEs. Additionally, let $o_{m,u}$ indicate that whether the UE u is located in cell m . If $o_{m,u} = 1$, the UE u is situated in cell m , and $o_{m,u} = 0$, otherwise.

The RMC connecting the gateway can collect the traffic demands and the channel status of UEs to make a BH time plan (BHTP) at each scheduling epoch including W timeslots. Generally, in order to reduce the complexity of BH scheduling, BHTP contains the scheduling results of T ($T \ll W$) consecutive timeslots, which is repeated in one scheduling epoch, as shown in Fig. 1. Denote the set of timeslots in BHTP as $\mathcal{T} = \{t \mid t = 1, 2, \dots, T\}$. Without loss of generality, we focus on the first S scheduling epochs. Let $\mathcal{S} = \{s \mid s = 1, 2, \dots, S\}$ be the set of scheduling epochs. Due to the short duration of each epoch, assume that the satellite-UE topology remains unchanged within one scheduling epoch but changes between two scheduling epochs [32]. Define $\xi_{n,u}^s$ to indicate that whether the satellite n is visible to the UE u during epoch s . If $\xi_{n,u}^s = 1$, the satellite n is within the Field of View (FoV) of UE u during epoch s , and $\xi_{n,u}^s = 0$, otherwise. The FoV is determined by the minimum elevation angle of UEs. Similarly, let $\gamma_{n,m}^s$ denote the visual relationship between cells and satellites. If the satellite n is within the FoV of cell m , $\gamma_{n,m}^s = 1$, and otherwise, $\gamma_{n,m}^s = 0$.

TABLE I
NOTATIONS AND DEFINITIONS

Notations	Definitions
$\mathcal{N}, N$	The set of satellites and the number of satellites
$\mathcal{N}_s^v$	The set of visible satellites by the service area in epoch s
$\mathcal{M}, M$	The set of cells and the number of cells
$\mathcal{M}_n^s$	The set of cells which contain UEs connected to satellite n at epoch s
$\mathcal{U}, U$	The set of UEs and the number of UEs
$\mathcal{U}_{n,m}^s$	The set of UEs connected with satellite n and located in cell m at epoch s
$\mathcal{U}_{n,t}^{s,t}$	The set of UEs belonging to the cells illuminated by satellite n at timeslot t in epoch s
$\mathcal{T}, T$	The set of timeslots in BHTP and the number of timeslots in BHTP
$\mathcal{S}, S$	The set of scheduling epochs and the number of scheduling epochs
$\mathcal{K}, K$	The set of spotbeams and the number of spotbeams
$\mathcal{F}, F$	The set of RBs and the number of RBs
B_w, B_{RB}	The bandwidth of the system and the bandwidth of each RB
σ^2	The noise power
T_{noise}	The noise temperature of the receiver
ϑ	The off-axis angle of UE
θ	The off-axis angle of satellite
$o_{m,u}$	Indication whether the UE u is situated in cell m
$\xi_{n,u}^s$	Indication whether the satellite n is visible to the UE u during epoch s
$\gamma_{n,m}^s$	Indication whether the satellite n is visible to the cell m during epoch s
$a_{n,u}^s$	The association relationship between UE u and satellite n during epoch s
$x_{n,k}^{s,m}(t)$	Indication whether the cell m is illuminated by satellite n 's spotbeam k at timeslot t in epoch s
$z_{n,k,f}^{s,u}(t)$	Indication whether the f -th RB of satellite n 's k -th spotbeam is assigned to UE u at timeslot t in epoch s
$p_{n,k,f}^{s,u}(t)$	The transmit power of UE u on spotbeam k 's RB f of satellite n at timeslot t in epoch s
d_u^s	The traffic demand of UE u in the s -th epoch
$d_{n,m}^s$	The traffic demand of cell m related to satellite n at epoch s
$h_{n,k,f}^{s,u}$	The channel coefficient between satellite n and UE u at the f -th RB of spotbeam k in epoch s
$\tilde{h}_{n,k}^{s,m}$	The average channel gain between cell m and satellite n 's spotbeam k in epoch s
$R_u^s(t)$	The achievable rate of UE u at timeslot t in epoch s
$\bar{R}_{n,m}^s$	The average achievable rate of cell m offered by satellite n at epoch s
$\hat{R}_u^s(t)$	The approximate achievable rate of UE u at timeslot t in epoch s
$\hat{R}_u^{s(i)}(t)$	The approximate achievable rate of UE u at timeslot t in epoch s during i -th SCA iteration
β_u^s	The service satisfaction level of UE u at epoch s
$\beta_u^s(t)$	The cumulative service satisfaction level of UE u at timeslot t in epoch s
Δ_u^s	Indication whether UE u execute an inter-satellite handover in epoch s
η_u^s	The actual transmission efficiency of UE u at epoch s
ASS_u^s	The actual service satisfaction of UE u at epoch s
$ASS_u^s(t)$	The cumulative actual service satisfaction of UE u at timeslot t in epoch s
$\hat{ASS}_u^s(t)$	The cumulative actual service satisfaction estimate of UE u at timeslot t in epoch s
$\hat{ASS}_u^{s(i)}(t)$	The approximate cumulative actual service satisfaction estimate of UE u at timeslot t in epoch s during i -th SCA iteration
D_{TH}, ϑ_{TH}	The interference distance threshold and the interference angle threshold
Y_n^s	The traffic load of satellite n at epoch s
$\omega_{n,j}^s(t)$	The dynamic weight of cell j in satellite n at timeslot t in epoch s
$G_{n,u}^{Tx}, G_{n,k,u}^{Rx}$	the transmit gain of UE u and the receiving gain of spotbeam k in satellite n
$r_{n,u}^s$	The distance between UE u and satellite n at epoch s

In the return-link BH system, each satellite can form at most K spotbeams to receive UEs' signals from different cells in a time-division multiple access (TDMA) manner. Denote $\mathcal{K} = \{k \mid k = 1, 2, \dots, K\}$ by the set of spotbeams. Specifically, each spotbeam can illuminate only one cell at one timeslot, and the set of cells illuminated by all spotbeams in a timeslot constitutes the BH pattern. UEs located in the illuminated cells can access to the satellite through RBs belonging to the spotbeam. Assume that each spotbeam has F orthogonal RBs and each spotbeam reuses the same system bandwidth B_w . Consequently, the bandwidth of each RB is $B_{RB} = B_w/F$ and the set of RBs is represented as $\mathcal{F} = \{f \mid f = 1, 2, \dots, F\}$.

Assume that each UE has only one antenna and the channel remains invariant within each epoch. Hence, the channel coefficient between satellite n and the u th UE at the f th RB of spotbeam k in epoch s can be modeled as

$$h_{n,k,f}^{s,u} = \sqrt{\frac{\xi_{n,u}^s}{\xi_{n,k,f}^{s,u} + 1}} \bar{g}_{n,k}^{s,u} + \sqrt{\frac{1}{\xi_{n,k,f}^{s,u} + 1}} \tilde{g}_{n,k,f}^{s,u} \quad (1)$$

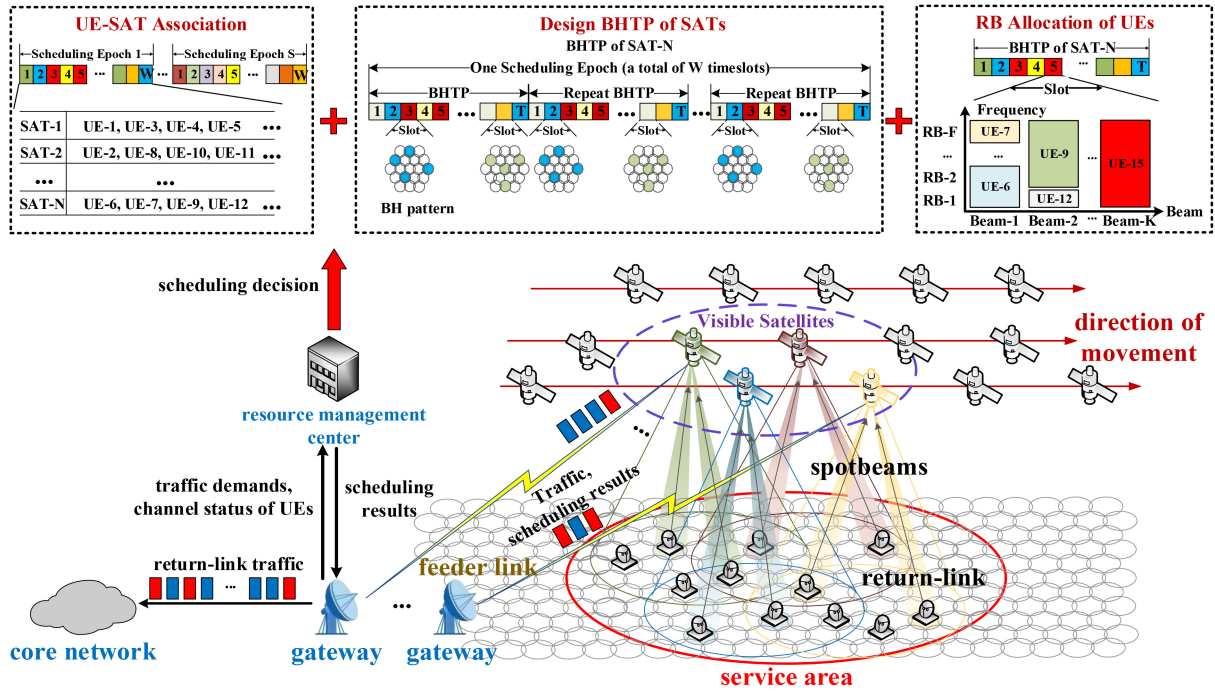


Fig. 1. System model of return-link BH scheduling for the NGSO mega-constellation.

where $\zeta_{n,k,f}^{s,u}$ denotes the Rice factor; $\bar{g}_{n,k}^{s,u}$ and $\tilde{g}_{n,k,f}^{s,u}$ represent the Line-of-Sight (LoS) and non-LoS components. Additionally, $|\bar{g}_{n,k}^{s,u}| = \sqrt{G_{n,u}^{Tx} G_{n,k,u}^{Rx} / (4\pi (r_{n,u}^s / \lambda)^2)}$ is determined by the free-space loss. $G_{n,u}^{Tx}$ and $G_{n,k,u}^{Rx}$ are the transmit gain of UE u and the receiving gain of spotbeam k in satellite n , respectively. $r_{n,u}^s$ represents the distance between UE u and satellite n at epoch s . λ is the wavelength of the carrier. The non-LoS component $\tilde{g}_{n,k,f}^{s,u}$ satisfies $\tilde{g}_{n,k,f}^{s,u} \sim \mathcal{CN}(0, |\bar{g}_{n,k}^{s,u}|^2)$.

Specifically, the UE antenna radiation pattern refers to ITU-R S.465-6 [33], which has the form as

$$G^{Tx}(\vartheta) = \begin{cases} G_{\max}^{Tx}, & \vartheta \leq \vartheta_{\min} \\ 32 - 25 \log_{10}(\vartheta), & \vartheta_{\min} < \vartheta \leq 48^\circ \\ -10, & 48^\circ < \vartheta \end{cases} \quad (2)$$

where ϑ denotes the off-axis angle in degrees and $\vartheta_{\min} = \max(2^\circ, 144(D/\lambda)^{-1.09})$. $G_{\max}^{Tx}$ represents the maximum gain of transmit antenna, which can be calculated by $G_{\max}^{Tx} = 10 * \log_{10}[(\pi D/\lambda)^2]$. D is the diameter of user antenna's aperture.

Besides, the satellite antenna radiation pattern refers to ITU-R S.1528 [34], which can be expressed as

$$G^{Rx}(\theta) = \begin{cases} G_{\max}^{Rx}, & \theta \leq \theta_b \\ G_{\max}^{Rx} - 3(\theta/\theta_b)^2, & \theta_b < \theta \leq Y \\ G_{\max}^{Rx} + L_s - 25 \log_{10}(\theta/Y), & Y < \theta \leq Z \\ L_F, & Z < \theta \end{cases} \quad (3)$$

where θ is the off-axis angle in degrees and θ_b denotes one half the 3-dB beamwidth θ_{3dB} . $G_{\max}^{Rx}$ is the maximum gain of receiving antenna, which can be denoted by $G_{\max}^{Rx} = 10 * \log_{10}[4.93(70/\theta_{3dB})^2]$. $Z = Y * 10^{0.04(G_{\max}^{Rx} + L_s - L_F)}$, where $L_s = -6.75$ (dB) represents the main beam and near-in side-lobe mask cross point below peak gain, $L_F = 0$ (dBi) denotes the far-out side-lobe level, and $Y = 1.5\theta_b$.

Define d_u^s as the traffic demand of UE u in the s th epoch. In each scheduling epoch, the RMC should make the on-demand resource management according to d_u^s and $h_{n,k,f}^{s,u}$. Denote $a_{n,u}^s \in \{0, 1\}$ by the association relationship between UE u and satellite n . $a_{n,u}^s = 1$ if UE u is assigned to satellite n in the s th epoch and $a_{n,u}^s = 0$, otherwise. Besides, the BH pattern of satellite n at timeslot t in epoch s is defined as $x_{n,k}^{s,m}(t) \in \{0, 1\}$. $x_{n,k}^{s,m}(t) = 1$ if the cell m is illuminated by satellite n 's spotbeam k and otherwise, $x_{n,k}^{s,m}(t) = 0$. Let $z_{n,k,f}^{s,u}(t) \in \{0, 1\}$ be the RB allocation variable. $z_{n,k,f}^{s,u}(t) = 1$ if the f th RB of satellite n 's k th spotbeam is assigned to UE u at timeslot t in epoch s and $z_{n,k,f}^{s,u}(t) = 0$ otherwise. Finally, we denote the transmit power of UE u on spotbeam k 's RB f of satellite n at timeslot t by $p_{n,k,f}^{s,u}(t)$. Therefore, the scheduling results of RMC at the s th epoch can be expressed as $\{a_{n,u}^s, x_{n,k}^{s,m}(t), z_{n,k,f}^{s,u}(t), p_{n,k,f}^{s,u}(t)\}$.

Assume that full frequency reuse between satellites and spotbeams, the achievable rate of UE u at timeslot t in epoch s can be represented as follows:

$$R_u^s(t) = \frac{B_{RB}}{T} \sum_{\substack{n \in \mathcal{N} \\ k \in \mathcal{K} \\ f \in \mathcal{F}}} \log_2 \left(1 + \frac{p_{n,k,f}^{s,u}(t) |h_{n,k,f}^{s,u}|^2}{\sum_{u' \in \mathcal{U} / u} p_{n,k,f}^{s,u'}(t) |h_{n,k,f}^{s,u'}|^2 + \sigma^2} \right) \quad (4)$$

where $\sigma^2 = \kappa B_{RB} T_{\text{noise}}$ denotes the noise power. κ and T_{noise} denote Boltzmann's constant and noise temperature of the receiver, respectively.

According to (4), we can use $\sum_{t \in \mathcal{T}} R_u^s(t)$ to represent the achievable rate of UE u in epoch s . Furthermore, denote $\beta_u^s = \sum_{t \in \mathcal{T}} R_u^s(t) / d_u^s$ by the service satisfaction level of UE u at epoch s , which is regarded as a gauge of the alignment between supply capacity and demand.

If the UE u is associated with different satellite at two consecutive epochs, the intersatellite handover occurs. In

particular, if $\sum_{n \in \mathcal{N}} a_{n,u}^{s-1} a_{n,u}^s = 0$, the UE u experiences the intersatellite handover from epoch $s-1$ to epoch s . Hence, let $\Delta_u^s = (1 - \sum_{n \in \mathcal{N}} a_{n,u}^{s-1} a_{n,u}^s)$ be the UE handover indicator variable. If $\Delta_u^s = 1$, UE u will execute an intersatellite handover in epoch s and $\Delta_u^s = 0$ otherwise. Considering that the handover process will introduce additional time overhead, the effective transmission capacity will be reduced. Thus, define $\eta_u^s = 1 - \Delta_u^s (T_{\text{handover}}/T_{\text{epoch}})$ as the actual transmission efficiency of UE u at epoch s with the consideration of handover overhead, where T_{handover} and T_{epoch} represent the duration of one handover and one epoch, respectively.

As a result, the actual service satisfaction (ASS) of UE u at epoch s can be represented as follows:

$$ASS_u^s = \min\{\eta_u^s \beta_u^s, 1\}. \quad (5)$$

Note that the ASS cannot exceed 1 since the actual transmission data volume for UEs should be less than the traffic requirements.

B. Problem Formulation

In this article, we formulate the return-link dynamic BH scheduling problem to the jointly optimize UE association, beam illumination and RB allocation with the objective of maximizing the long-term average ASS of UEs, which can be expressed as

$$\mathcal{P}_0: \max_{\{a_{n,u}^s, x_{n,k}^{s,m}(t), z_{n,k,f}^{s,u}(t), p_{n,k,f}^{s,u}(t)\}} \frac{\sum_{s \in \mathcal{S}, u \in \mathcal{U}} ASS_u^s}{U \cdot S} \quad (6a)$$

$$\text{s.t. } C_1: a_{n,u}^s \leq \xi_{n,u}^s \quad \forall n, u, s \quad (6b)$$

$$C_2: \sum_{n \in \mathcal{N}} a_{n,u}^s \leq 1 \quad \forall u, s \quad (6c)$$

$$C_3: x_{n,k}^{s,m}(t) \leq \gamma_{n,m}^s \quad \forall n, m, s, k, t \quad (6d)$$

$$C_4: \sum_{m \in \mathcal{M}, k \in \mathcal{K}} x_{n,k}^{s,m}(t) \leq K \quad \forall n, s, t \quad (6e)$$

$$C_5: \sum_{m \in \mathcal{M}} x_{n,k}^{s,m}(t) \leq 1 \quad \forall n, k, s, t \quad (6f)$$

$$C_6: z_{n,k,f}^{s,u}(t) \leq \sum_{m \in \mathcal{M}} x_{n,k}^{s,m}(t) a_{n,u}^s o_{m,u} \quad \forall n, k, f, u, s, t \quad (6g)$$

$$C_7: \sum_{f \in \mathcal{F}, u \in \mathcal{U}} z_{n,k,f}^{s,u}(t) \leq F \quad \forall n, k, s, t \quad (6h)$$

$$C_8: \sum_{u \in \mathcal{U}} z_{n,k,f}^{s,u}(t) \leq 1 \quad \forall n, k, f, s, t \quad (6i)$$

$$C_9: 0 \leq p_{n,k,f}^{s,u}(t) \leq z_{n,k,f}^{s,u}(t) P_{\max} \quad \forall n, k, f, s, u, t \quad (6j)$$

$$C_{10}: \sum_{n \in \mathcal{N}, k \in \mathcal{K}, f \in \mathcal{F}} p_{n,k,f}^{s,u}(t) \leq P_{\max} \quad \forall u, s, t \quad (6k)$$

$$C_{11}: a_{n,u}^s, x_{n,k}^{s,m}(t), z_{n,k,f}^{s,u}(t) \in \{0, 1\} \quad \forall m, n, k, f, s, u, t \quad (6l)$$

where $\{a_{n,u}^s, x_{n,k}^{s,m}(t), z_{n,k,f}^{s,u}(t), p_{n,k,f}^{s,u}(t)\}$ are the decision variables to be optimized. Constraints C_1 and C_2 mean that UEs can only connect to one visual satellite at most. Constraint C_3 indicates that cells have to be illuminated by the satellites within the FoV. Constraint C_4 constrains the maximum number of spotbeams per satellite. In C_5 , each spotbeam can illuminate

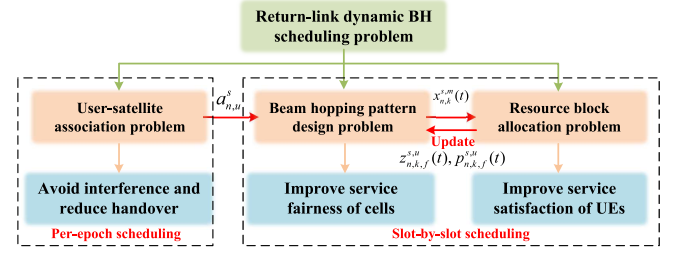


Fig. 2. Architecture of problem decomposition.

one cell at most. Constraint C_6 indicates that the UE can use the RBs of the spotbeam that is illuminating to him while the spotbeam must belong to the accessed satellite. Constraint C_7 constrains the maximum number of RBs per spotbeam and C_8 ensures that one RB can only be allocated to one UE. Constraints C_9 and C_{10} are the power constraints of UEs, where $P_{\max}$ denotes the maximum transmit power.

Note that the optimization problem $\mathcal{P}_0$ is a nonlinear and nonconvex mixed-integer optimization problem, which is NP-hard. Consequently, finding the optimal solution is exceptionally challenging under affordable complexity. This article will propose a low-complexity algorithm to obtain a suboptimal solution in the next section.

III. PROBLEM DECOMPOSITION AND SOLVING METHOD

In order to make the problem $\mathcal{P}_0$ tractable, this article decouples this problem into three subproblems, namely, user-satellite association problem, BH pattern design problem, and RB allocation problem. As shown in Fig. 2, first, the user-satellite association problem is solved to determine the connection between UEs and satellites per scheduling epoch (i.e., $a_{n,u}^s$). Next, BH pattern design problem will decide the beam illumination of each satellite at each timeslot within the BHTP [i.e., $x_{n,k}^{s,m}(t)$] under the given $a_{n,u}^s$. Finally, according to $a_{n,u}^s$ and $x_{n,k}^{s,m}(t)$, RB allocation problem will be solved to obtain the RB allocation to UEs [i.e., $z_{n,k,f}^{s,u}(t)$ and $p_{n,k,f}^{s,u}(t)$] slot-by-slot.

A. User-Satellite Association Problem and Solution

When assigning UEs to satellites, the following factors need to be considered.

1) *Avoiding Intersatellite Interference:* In order to mitigate intersatellite interference, we first analyze the generation mechanism of the interference. As depicted in Fig. 3, the uplink intersatellite interference cannot be ignored when the interference off-axis angle (i.e., θ and ϑ) are both very small, so that the interference will be emitted by the main lobe of the UE antenna and received by the main lobe of the satellite antenna. Obviously, this scenario occurs when two adjacent UEs connect to two close satellites. This article assumes that two UEs are adjacent if their distance is less than $2 \times r_{\text{cell}}$, where r_{cell} represents the radius of the cell. Besides, we define the interference distance threshold D_{TH} to determine whether two satellites are close. For example, if the distance between satellite i and satellite j , denoted by $D_{i,j}$, is less than D_{TH} , these two satellites are adjacent so that two adjacent UEs cannot be

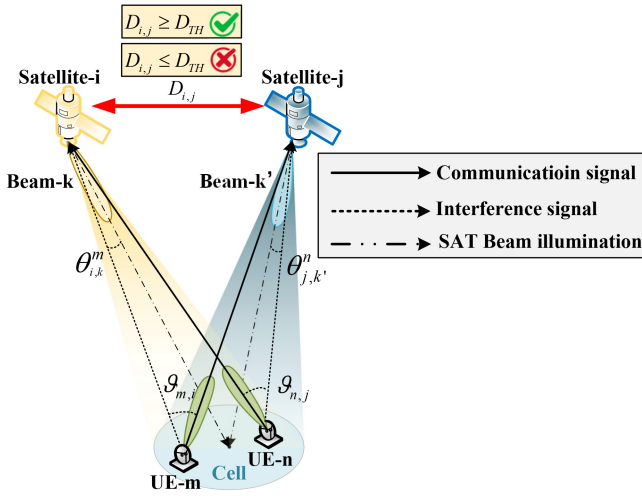


Fig. 3. Illustration of the uplink intersatellite interference.

assigned to them simultaneously. Specifically, the interference distance threshold D_{TH} is expressed as follows:

$$D_{TH} = \max\{\sqrt{2(1 - \cos(\vartheta_{TH}))}r_{\max}, \sqrt{r_{\max}^2 + r_{\min}^2 - 2r_{\max}r_{\min}\cos(\vartheta_{TH})}\}. \quad (7)$$

$r_{\max}$ and $r_{\min}$ denote the maximum and minimum distance from UE to visible satellites, which can be calculated by $r_{\max} = \sqrt{(r_e \sin(\alpha_{\min}))^2 + H^2 + 2r_e H - r_e \sin(\alpha_{\min})}$ and $r_{\min} = r_e + H$, respectively. Particularly, r_e , H , and $\alpha_{\min}$ represent the Earth radius, satellite height, and minimum user elevation angle, respectively. Furthermore, ϑ_{TH} is a preset interference angle threshold, which means the expected off-axis angle of the interference transmitted by UE antenna.

Proposition 1: If the distance between visible satellite i and visible satellite j satisfies $D_{i,j} > D_{TH}$, the angle between arbitrary UE and these two satellites ϑ is larger than ϑ_{TH} .

Proof: Refer to the Appendix. ■

According to Proposition 1, the uplink interference can be controlled under a certain level if $D_{i,j} > D_{TH}$. Note that although adjacent UEs cannot access adjacent satellites, they could access the same satellite because this satellite can avoid interuser interference by designing reasonable BH patterns.

2) *Balancing Intersatellite Traffic Load:* Denote Y_n^s as the traffic load of satellite n at epoch s . Then, we can calculate the traffic load by $Y_n^s = \sum_{u \in \mathcal{U}} d_u^s a_{n,u}^s$. If some satellites are overloaded or unloaded, the resources of satellites will be unable to meet UEs' traffic demands or be wasted. Therefore, it is necessary to balance the traffic load of each satellite as much as possible when associating UEs to satellites.

3) *Reducing Intersatellite Handover:* In order to improve the ASS of UEs defined in (5), we should avoid frequent intersatellite handover. As a result, when making the decision of UE-satellite association for the next epoch, the last UE-satellite connection results need to be considered.

Based on the above considerations, this article designs a low-complexity heuristic user-satellite association strategy, which is described in Algorithm 1. First, we sort UEs

Algorithm 1 Low-Complexity Heuristic User-Satellite Association Strategy

Input: The traffic demand of UEs d_u^s ; the position of UEs and satellites; visible relationship between UEs and satellites $\xi_{n,u}^s$; user-satellite association results in last epoch $a_{n,u}^{s-1}$; the interference angle threshold ϑ_{TH} .

Initialize: The set of accessed UEs $\mathcal{A} = \emptyset$; the traffic load of satellites $Y_n^s = 0 \forall n \in \mathcal{N}$; the user-satellite association results $a_{n,u}^s = 0 \forall n \in \mathcal{N} \forall u \in \mathcal{U}$.

- 1: Sort UEs in descending order according to traffic demands: $(u_1, u_2, \dots, u_U) | d_{u_1}^s \geq d_{u_2}^s \geq \dots \geq d_{u_U}^s$.
- 2: **for** $u = u_1$ to u_U **do**
- 3: Find the set of adjacent users of UE u from set $\mathcal{A}$: π_u .
- 4: Obtain the set of satellites accessed by UEs in π_u : v_u .
- 5: Obtain the set of visible satellites of UE u : Υ_u .
- 6: Initialize UE u 's candidate satellite set: $\Omega_u = \emptyset$.
- 7: **repeat**
- 8: Find the satellites from Υ_u which interfere with the satellites in v_u , according to the interference distance (7): $\Psi_u = \{n | D_{n,m} \leq D_{TH}, n \in \Upsilon_u, m \in v_u\}$.
- 9: Calculate the candidate satellites for UE u : $\Omega_u = \Upsilon_u - \Psi_u$.
- 10: Reduce the interference angle threshold: $\vartheta_{TH} = \vartheta_{TH} - 5^\circ$.
- 11: **until** $\Omega_u \neq \emptyset$
- 12: **if** $a_{n^*,u}^{s-1} = 1$ **and** $n^* \in \Omega_u$ **then**
- 13: Assign UE u to satellite n^* : $a_{n^*,u}^s = 1$.
- 14: **else**
- 15: Find the satellite with the minimum traffic load from the candidate set Ω_u : $n^* = \arg \min_{n \in \Omega_u} Y_n^s$.
- 16: Assign UE u to satellite n^* : $a_{n^*,u}^s = 1$.
- 17: **end if**
- 18: Update the set of accessed UEs: $\mathcal{A} = \mathcal{A} \cup u$.
- 19: Update the traffic load of satellite n^* : $Y_{n^*}^s = Y_{n^*}^s + d_u^s$.
- 20: **end for**

Output: The user-satellite association results: $\{a_{n,u}^s\}$.

in descending order according to traffic demands (line 1). Then, we select satellite for UEs one by one in this order (lines 2–20). In particular, in terms of UE u 's satellite association, we first find the users who have completed association and are adjacent to UE u (denoted by π_u), and obtain the satellites accessed by these users (denoted by v_u) (lines 3 and 4). Next, the satellites Ψ_u which may interfere with v_u can be found according to the interference distance D_{TH} , and finally the candidate satellites for UE u can be obtained by removing interfering satellites from the visible satellites of UE u (lines 5–11). Considering that the candidate satellites set may be empty if the interference angle threshold ϑ_{TH} is set too large, we need to gradually reduce ϑ_{TH} until the candidate set is obtained. In order to reduce handover overhead, if the candidate set contains the last accessed satellite, we assign UE u to this satellite (lines 12 and 13). Otherwise, we associate UE u with the satellite which has the minimum traffic load from the candidate set, for the purpose of balancing traffic load

among satellites (lines 14–17). Eventually, we can obtain the user–satellite association results $\{a_{n,u}^s\}$ of scheduling epoch s .

B. Beam Hopping Pattern Design Problem and Solution

Given the user–satellite association results of epoch s , BH patterns of each satellite will be designed. In other words, we should determine the beam illumination direction of each satellite at each timeslot within the epoch s (i.e., $\{x_{n,k}^{s,m}(t) \forall n, k, m\}$). Considering that designing a joint multisatellite BH pattern is much complex, this article independently designs the BH pattern for each satellite since the intersatellite interference has been effectively avoided when associating UEs to satellites. Here, we regard each cell as a virtual UE, which aggregates the traffic demands of all access UEs located in this cell. Denote $\mathcal{U}_{n,m}^s = \{u | o_{m,u} a_{u,n}^s = 1, u \in \mathcal{U}\}$ by the set of UEs associated with satellite n and located in cell m at epoch s . Accordingly, the demand of cell m (or virtual UE m) related to satellite n at epoch s can be obtained as $d_{n,m}^s = \sum_{u \in \mathcal{U}_{n,m}^s} d_u^s$. Similarly, the average channel gain between cell m and satellite n 's spotbeam k can be expressed as $|\tilde{h}_{n,k}^{s,m}| = \text{mean}_{u \in \mathcal{U}_{n,m}^s} (|h_{n,k}^{s,u}|)$. Therefore, according to the BH patterns $\{x_{n,k}^{s,m}(t)\}$ and the average channel gain $|\tilde{h}_{n,k}^{s,m}|$, we can estimate the average capacity of cell m offered by satellite n at epoch s as follows:

$$\tilde{R}_{n,m}^s = \frac{B_w}{T} \sum_{k \in \mathcal{K}, t \in \mathcal{T}} \log_2 \left(1 + \frac{x_{n,k}^{s,m}(t) P_{\max} |\tilde{h}_{n,k}^{s,m}|^2}{\sum_{m' \in \mathcal{M}/m} x_{n,k}^{s,m'}(t) P_{\max} |\tilde{h}_{n,k}^{s,m'}|^2 + \tilde{\sigma}^2} \right) \quad (8)$$

where $\tilde{\sigma}^2 = \kappa B_w T_{\text{noise}}$ represents the total noise power on the whole bandwidth.

In order to achieve on-demand services and ensure fairness among cells, the BH pattern design problem of satellite n at epoch s is formulated to maximize the minimum service satisfaction of cells that satellite n serves, which has the form as

$$\mathcal{P}_1: \max_{\{x_{n,k}^{s,m}(t) \forall k, m, t\}} \min_{m \in \mathcal{M}_n^s} \left\{ \frac{\tilde{R}_{n,m}^s}{d_{n,m}^s} \right\} \quad (9a)$$

$$\text{s.t.} \quad (6d) - (6f), (6l) \quad (9b)$$

where $\mathcal{M}_n^s = \{m | \mathcal{U}_{n,m}^s \neq \emptyset, m \in \mathcal{M}\}$ represents the set of cells that contain UEs connected to satellite n at epoch s .

Note that $\mathcal{P}_1$ is still a nonconvex integer optimization problem due to the nonconvex of (8). Additionally, $\mathcal{P}_1$ requires solving all BH patterns in the whole BHTP, resulting in huge computational complexity. To make $\mathcal{P}_1$ tractable, we propose a maximum dynamic weight (MDW) method to decide the BH pattern slot-by-slot. First, we build a weighted interference graph $\mathcal{G}(\mathcal{V}_n^s, \mathcal{E}_n^s)$ for satellite n , where $\mathcal{V}_n^s$ and $\mathcal{E}_n^s$ are the set of vertices and edges, respectively. Specifically, vertex $v_{n,j}^s \in \mathcal{V}_n^s$ represents the j th element of $\mathcal{M}_n^s$ (i.e., the j th cell that satellite n associate with at epoch s). Besides, the attribute of $v_{n,j}^s$ is the dynamic weight of cell j at timeslot t , denoted by $\omega_{n,j}^s(t)$. Specifically, the dynamic weight $\omega_{n,j}^s(t)$ has the form as

$$\omega_{n,j}^s(t) = - \min_{u \in \mathcal{U}_{n,v_{n,j}^s}^s} \{\beta_u^s(t-1)\} \quad (10)$$

where $\beta_u^s(t-1) = \sum_{i=1}^{t-1} R_u^s(i)/d_u^s$ represents the cumulative service satisfaction level of UE u at timeslot $t-1$ in epoch s

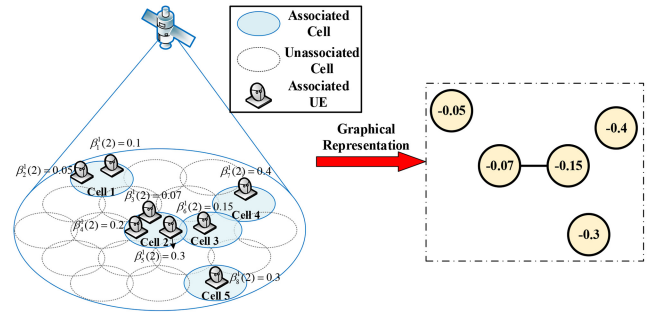


Fig. 4. Illustration of the weighted interference graph with five associated cells and eight associated UEs ($s = 1$, $t = 3$, and $F = 3$).

and $R_u^s(t)$ is defined as (4). Note that the weight can be updated every timeslot, and this update requires the BH pattern and the resource allocation results of previous timeslots. The resource allocation method will be detailed in the next section.

Additionally, we use edges to indicate the potential interference between cells. There exists an edge between two vertices if the following two interference conditions are satisfied simultaneously.

- 1) Two vertices represent adjacent cells.
- 2) The sum of the number of UEs in the cells represented by the two vertices exceeds F (i.e., the number of RBs).

Fig. 4 illustrates an example at the third timeslot of the first scheduling epoch with $F = 3$. There are five associated cells and eight associated UEs. Since cell 2 and cell 3 are adjacent, and the total number of UEs in these cells exceeds 3, there exists an edge between them.

To obtain the BH patterns, we should find at most K vertices from the weighted interference graph at each timeslot such that any two vertices are not connected and the sum of weights is maximum. We propose a greedy algorithm to search vertices for each timeslot, as shown in Algorithm 2. At first, we select the vertex with maximum weight to be illuminated by the spotbeam (lines 2–4). Second, remove this vertex and its adjacent vertices from the vertex set (lines 5 and 6). Then, repeat above steps until K vertices have been selected or the vertex set is empty. Finally, we can obtain the BH pattern of satellite n at timeslot t .

C. Resource Block Allocation Problem and Solution

In this section, the RB allocation will be optimized under the given user–satellite association and the BH pattern results. Similar to the last section, we solve this problem slot by slot for each satellite to reduce the computational complexity. The RB allocation problem of satellite n at timeslot t can be modeled as maximizing the average ASS of UEs in the illuminated cells, which is formulated as follows:

$$\mathcal{P}_2: \max_{\mathbf{z}_n^s(t), \mathbf{p}_n^s(t)} \frac{\sum_{u \in \mathcal{U}_n^{s,t}} \hat{A} \hat{S}_u^s(t)}{|\mathcal{U}_n^{s,t}|} \quad (11a)$$

$$\text{s.t.} \quad (6g) - (6l) \quad (11b)$$

where $\mathbf{z}_n^s(t) \triangleq [z_{n,k,f}^{s,u}(t)] \forall u, k, f$; $\mathbf{p}_n^s(t) \triangleq [p_{n,k,f}^{s,u}(t)] \forall u, k, f$; $\mathcal{U}_n^{s,t} = \bigcup_{m \in \{m | x_{n,k}^{s,m}(t)=1\}} \mathcal{U}_{n,m}^s$ denotes the set of UEs belonging to the cells illuminated by satellite n at timeslot t in epoch s .

Algorithm 2 Greedy Algorithm Based on Maximum Weight Search

Input: The weighted interference graph $\mathcal{G}(\mathcal{V}_n^s, \mathcal{E}_n^s)$ at timeslot t in epoch s .

Initialize: $k = 1, \mathcal{H} = \mathcal{V}_n^s$.

- 1: **repeat**
- 2: Select the vertex with maximum weight from $\mathcal{H}$: $v_{n,j}^s = \arg \max_{v_{n,j}^s \in \mathcal{H}} \omega_{n,j}^s(t)$.
- 3: Obtain the cell m^* corresponding to the vertex $v_{n,j}^s$.
- 4: Spotbeam k of satellite n illuminates the cell m^* at timeslot t : $x_{n,k}^{s,m^*}(t) = 1$.
- 5: Find the set of vertices adjacent to $v_{n,j}^s$ from $\mathcal{H}$: Γ .
- 6: Update the vertex set: $\mathcal{H} = \mathcal{H} - \Gamma - \{v_{n,j}^s\}$.
- 7: $k = k + 1$.
- 8: **until** $k > K$ or $\mathcal{H} = \emptyset$

Output: The BH pattern of satellite n at timeslot t : $\{x_{n,k}^{s,m}(t)\}$.

Besides, $\hat{A}SS_u^s(t)$ represents the cumulative ASS estimate of UE u at timeslot t , which can be calculated sequentially by

$$\hat{A}SS_u^s(t) = \min \left\{ ASS_u^s(t-1) + \frac{\eta_u \hat{R}_u^s(t)}{d_u^s}, 1 \right\} \quad (12)$$

where $ASS_u^s(t-1) = \min\{(\eta_u \sum_{i=1}^{t-1} R_u^s(t)/d_u^s), 1\}$ is the actual cumulative ASS of UE u at timeslot $t-1$, known at timeslot t . Furthermore, $\hat{R}_u^s(t)$ is the approximate achievable rate of UE u at timeslot t in epoch s assuming that the intersatellite interference has been avoided, which can be represented as

$$\hat{R}_u^s(t) = \frac{B_{RB}}{T} \sum_{k \in \mathcal{K}, f \in \mathcal{F}} \log_2 \left(1 + \frac{p_{n,k,f}^{s,u}(t) |h_{n,k,f}^{s,u}|^2}{\sum_{u' \in \mathcal{U}_n^{s,t}/u} p_{n,k,f}^{s,u'}(t) |h_{n,k,f}^{s,u'}|^2 + \sigma^2} \right). \quad (13)$$

It can be observed that $\mathcal{P}_2$ is a mixed integer nonconvex programming, which is NP-hard. This article will transform $\mathcal{P}_2$ into a more tractable form and use SCA-based method [35], [36], [37] to solve it. At first, we relax the binary matrix $\mathbf{z}_n^s(t)$ into continuous one (i.e., $0 \leq z_{n,k,f}^{s,u}(t) \leq 1 \forall u, k, f$). Then, a penalty function is defined to accelerate the convergence of SCA [38], [39], which can be expressed as $\Lambda(\mathbf{z}_n^s(t)) = \sum_{u,k,f} [(z_{n,k,f}^{s,u}(t))^2 - z_{n,k,f}^{s,u}(t)]$. By introducing $\Lambda(\mathbf{z}_n^s(t))$ into the objective function (11a), $\mathcal{P}_2$ can be renewed as

$$\mathcal{P}_3: \max_{\mathbf{z}_n^s(t), \mathbf{p}_n^s(t)} \frac{\sum_{u \in \mathcal{U}_n^{s,t}} \hat{A}SS_u^s(t)}{|\mathcal{U}_n^{s,t}|} + \mu \Lambda(\mathbf{z}_n^s(t)) \quad (14a)$$

$$\text{s.t.} \quad (6g) - (6k), \quad (14b)$$

$$0 \leq z_{n,k,f}^{s,u}(t) \leq 1 \forall u, k, f \quad (14c)$$

where $\mu > 0$ is a penalty parameter.

Proposition 2: Given a suitable large value of μ , $\mathcal{P}_2$ and $\mathcal{P}_3$ have the same optimal solution.

The proof of Proposition 2 can be similarly followed by [40] and [41], which show that the optimal solution is obtained when $\Lambda(\mathbf{z}_n^s(t)) = 0$ and implies that $\mathbf{z}_n^s(t)$ are binary at optimum if μ is large.

Note that the objective function in $\mathcal{P}_3$ is nonconcave due to $\hat{R}_u^s(t)$ (in $\hat{A}SS_u^s(t)$) and $\Lambda(\mathbf{z}_n^s(t))$. Let

Algorithm 3 SCA-Based Iterative Algorithm to Solve $\mathcal{P}_3$

Input: The maximum number of iterations I_{max} .

Initialize: Set $i = 1$ and give the initial feasible points $(\mathbf{z}_n^{s(0)}(t), \mathbf{p}_n^{s(0)}(t))$.

- 1: **repeat**
- 2: Solve $\mathcal{P}_4$ to obtain $(\mathbf{z}_n^{s*}(t), \mathbf{p}_n^{s*}(t))$ and Θ^* .
- 3: Update $(\mathbf{z}_n^{s(i)}(t), \mathbf{p}_n^{s(i)}(t)) := (\mathbf{z}_n^{s*}(t), \mathbf{p}_n^{s*}(t))$, $\Theta^{(i)} := \Theta^*$.
- 4: $i = i + 1$.
- 5: **until** $i > I_{max}$
- 6: Recover the binary solution $\mathbf{z}_n^{s*}(t) = \lfloor \mathbf{z}_n^{s*}(t) + 0.5 \rfloor$.

Output: $\{\mathbf{z}_n^{s*}(t), \mathbf{p}_n^{s*}(t)\}$.

$\hat{R}_u^s(t) = L(\mathbf{p}_n^s(t)) - V(\mathbf{p}_n^s(t))$, where $L(\mathbf{p}_n^s(t)) = (B_{RB}/T) \sum_{k \in \mathcal{K}, f \in \mathcal{F}} \log_2(\sum_{u' \in \mathcal{U}_n^{s,t}} p_{n,k,f}^{s,u'}(t) |h_{n,k,f}^{s,u'}|^2 + \sigma^2)$ and $V(\mathbf{p}_n^s(t)) = (B_{RB}/T) \sum_{k \in \mathcal{K}, f \in \mathcal{F}} \log_2(\sum_{u' \in \mathcal{U}_n^{s,t}/u} p_{n,k,f}^{s,u'}(t) |h_{n,k,f}^{s,u'}|^2 + \sigma^2)$. Since $\hat{R}_u^s(t)$ is a difference of concave function, we use first-order Taylor approximation to linearize the term $V(\mathbf{p}_n^s(t))$ at the i th SCA iteration as follows:

$$\tilde{V}^{(i)}(\mathbf{p}_n^s(t)) = V(\mathbf{p}_n^{s(i-1)}(t)) + \nabla V_{\mathbf{p}_n^s(t)}(\mathbf{p}_n^{s(i-1)}(t))(\mathbf{p}_n^s(t) - \mathbf{p}_n^{s(i-1)}(t)) \quad (15)$$

where $\mathbf{p}_n^{s(i-1)}(t)$ represents the $(i-1)$ th SCA iteration result.

Thus, $\hat{R}_u^s(t)$ can be approximated as $\tilde{R}_u^{s(i)}(t) = L(\mathbf{p}_n^s(t)) - \tilde{V}^{(i)}(\mathbf{p}_n^s(t))$ at the i th iteration, which is concave. Meanwhile, the approximation of $\hat{A}SS_u^s(t)$ can be expressed as follows:

$$\tilde{A}SS_u^{s(i)}(t) = \min \left\{ ASS_u^s(t-1) + \frac{\eta_u \tilde{R}_u^{s(i)}(t)}{d_u^s}, 1 \right\}. \quad (16)$$

In addition, we can also linearize $\Lambda(\mathbf{z}_n^s(t))$ using first-order Taylor approximation at the i th SCA iteration as follows:

$$\tilde{\Lambda}^{(i)}(\mathbf{z}_n^s(t)) = \sum_{u,k,f} \left[\left(z_{n,k,f}^{s,u(i-1)}(t) \right)^2 + 2z_{n,k,f}^{s,u(i-1)}(t)[z_{n,k,f}^{s,u}(t) - z_{n,k,f}^{s,u(i-1)}(t)] - z_{n,k,f}^{s,u}(t) \right] \quad (17)$$

where $z_{n,k,f}^{s,u(i-1)}(t)$ represents the $(i-1)$ th SCA iteration result. Based on the aforementioned approximations (16) and (17), the convex approximate program of $\mathcal{P}_3$ at the i th SCA iteration can be stated as follows:

$$\mathcal{P}_4: \max_{\mathbf{z}_n^s(t), \mathbf{p}_n^s(t)} \Theta^{(i)} \triangleq \frac{\sum_{u \in \mathcal{U}_n^{s,t}} \tilde{A}SS_u^{s(i)}(t)}{|\mathcal{U}_n^{s,t}|} + \mu \tilde{\Lambda}^{(i)}(\mathbf{z}_n^s(t)) \quad (18a)$$

$$\text{s.t.} \quad (6g) - (6k), (14c). \quad (18b)$$

$\mathcal{P}_4$ can be effectively solved using interior point method by CVX toolbox [42]. We summarize the SCA-based method in Algorithm 3. Lines 1–5 describe the iteration process to refine the solution and line 6 recovers the binary variable to ensure a feasible solution.

Finally, after solving above three subproblems, we can obtain the whole BH scheduling results. The overall framework of proposed algorithm to solve the original optimization problem $\mathcal{P}_0$ is summarized in Algorithm 4.

Algorithm 4 Overall Algorithm Framework to Solve $\mathcal{P}_0$ **Input:** $N, S, T, F, K, B_w, d_u^s, h_{n,k,f}^{s,u}, O_{m,u}, \gamma_{n,u}^s, \xi_{n,u}^s, \theta_{TH}, I_{max}$.**Initialize:** Set $a_{n,u}^0 = 0 \forall n, u$.

```

1: for  $s = 1$  to  $S$  do
2:   /*User-Satellite Association*/
3:   Obtain the user-satellite association result at epoch  $s$ 
      $\{a_{n,u}^s\}$  using Algorithm 1.
4:   for  $t = 1$  to  $T$  do
5:     for  $n = 1$  to  $N$  do
6:       /*Beam Hopping Pattern Design*/
7:       Obtain the weighted interference graph  $\mathcal{G}(\mathcal{V}_n^s, \mathcal{E}_n^s)$ 
         of satellite  $n$ .
8:       Calculate the vertex weight  $\omega_{n,j}^s(t)$  using (10).
9:       Obtain the BH pattern of satellite  $n$  at timeslot  $t$ 
          $\{x_{n,k}^{s,m}(t)\}$  using Algorithm 2.
10:      /*Resource Block Allocation*/
11:      Obtain the RB allocation result of satellite  $n$  at
         timeslot  $t$   $\{z_n^s(t), \mathbf{p}_n^s(t)\}$  using Algorithm 3.
12:    end for
13:    Calculate  $R_u^s(t)$  according to (4) and obtain  $ASS_u^s(t)$ .
14:  end for
15: end for
Output: Scheduling results:  $\{a_{n,u}^s, x_{n,k}^{s,m}(t), \mathbf{z}_n^s(t), \mathbf{p}_n^s(t)\}$ .

```

TABLE II
SYSTEM PARAMETERS

PARAMETERS	VALUES
Satellite altitude H	550 km
The number of satellite orbits	72
The number of satellites per orbit	44
The number of satellites N	3168
Orbit inclination	53°
The center location of the specific area	(76°W, 40°N)
The number of cells in the specific area M	100
The radius of a cell r_{cell}	15 km
Uplink frequency	14.25 GHz
Satellite 3 dB beamwidth θ_{3dB}	3°
Maximum receiving antenna gain G_{max}^{Rx}	34.3 dBi
Maximum number of spotbeams per satellite K	8
System bandwidth B_w	250 MHz
The number of RBs F	5
The number of UEs U	50 ~ 700
The traffic demand of UEs d_u^s	1 ~ 50 Mbps
The diameter of user antenna's aperture D	0.4 m
Maximum transmit antenna gain G_{max}^{Tx}	32.5 dBi
Maximum transmit power of UE P_{max}	2 Watt
Noise temperature T_{noise}	300 K
Rice factor $\kappa_{n,k,f}^{s,u}$	10 dB
Interference angle threshold ϑ_{TH}	20°
Minimum user elevation angle α_{min}	40° [43]
Maximum number of SCA iterations I_{max}	3
The number of timeslots in BHTP T	15
The number of scheduling epochs S	60
The duration of a timeslot	2 ms
The duration of a scheduling epoch T_{epoch}	15 s
The handover time overhead $T_{handover}$	3 s
Simulation duration	15 min

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

A. Simulation Parameter

In this article, a LEO mega constellation is considered, where a total of 3168 satellites are uniformly distributed across 72 orbits with a 53° inclination at an altitude of 550 km. Table II summarizes the system parameters. Some of them refer to Starlink [43], [44]. In order to simulate dynamic satellite topology, the simulation lasts for 15 min and each scheduling epoch is set to 15 s. Each satellite has at most 8 Ku-band uplink spotbeams which share a bandwidth of 250 MHz. The number of RBs in each spotbeam is set as 5. The specific area consists of 100 cells, the center of which is located in (76°W, 40°N). In order to reflect the nonuniform and time-varying traffic, the traffic demands of UEs are sampled from 1 to 50 Mb/s and vary between different scheduling epochs. Besides, we consider two scenarios for user geographic distribution. One follows a uniform distribution, and the other follows a nonuniform distribution based on actual world population density [45]. The minimum elevation angle of UEs is set to 40° while the interference angle threshold is set to 20°. Besides, the handover time overhead is set to 3 s. Additionally, the BHTP contains 15 timeslots and the maximum transmit power of UE is set as 2 W.

B. Performance Analysis

Three benchmarks are compared to the proposed user-satellite association algorithm.

- 1) *BestChannel*: UEs are associated with the satellite with the best channel gains at each scheduling epoch.
- 2) *MaxElevation* [46]: UEs are assigned to the satellite which has the maximum elevation angle at each scheduling epoch.

3) *MaxTime*: UEs are allocated to the satellite that provides the maximum service duration at each scheduling epoch. Besides, we compare the proposed BH pattern design method (denoted by MDW-BH) with three baselines.

- 1) *Random-BH*: In each timeslot, the satellite randomly selects K access cells to serve.
- 2) *Round-Robin-BH*: The access cells will be periodically illuminated by the satellite's spotbeams following the Round-Robin basis.
- 3) *Greedy-BH*: In each timeslot, the satellite selects the K access cells with the lowest service satisfaction to illuminate.

Meanwhile, the proposed SCA-based RB allocation scheme will be compared to the *EvenPower* baseline, where RBs are allocated to the UEs with minimum cumulative ASS and the transmit power of the UE is evenly distributed on allocated RBs at each timeslot.

This article selects the following four performance metrics to evaluate algorithms.

- 1) *Average ASS of UEs*: This performance metric characterizes the satellite's on-demand service capability for users, which is the optimization target of $\mathcal{P}_0$.
- 2) *Average Fairness of UEs*: We use Jain's fairness index to measure the service fairness among UEs [47], which can be expressed as $\text{mean}_{s \in S} ((\sum_{u \in \mathcal{U}} ASS_u^s)^2 / U \sum_{u \in \mathcal{U}} (ASS_u^s)^2)$.
- 3) *The Worst UE's Service Satisfaction*: This metric denotes the lowest service satisfaction of UEs and can represent the worst case of service in

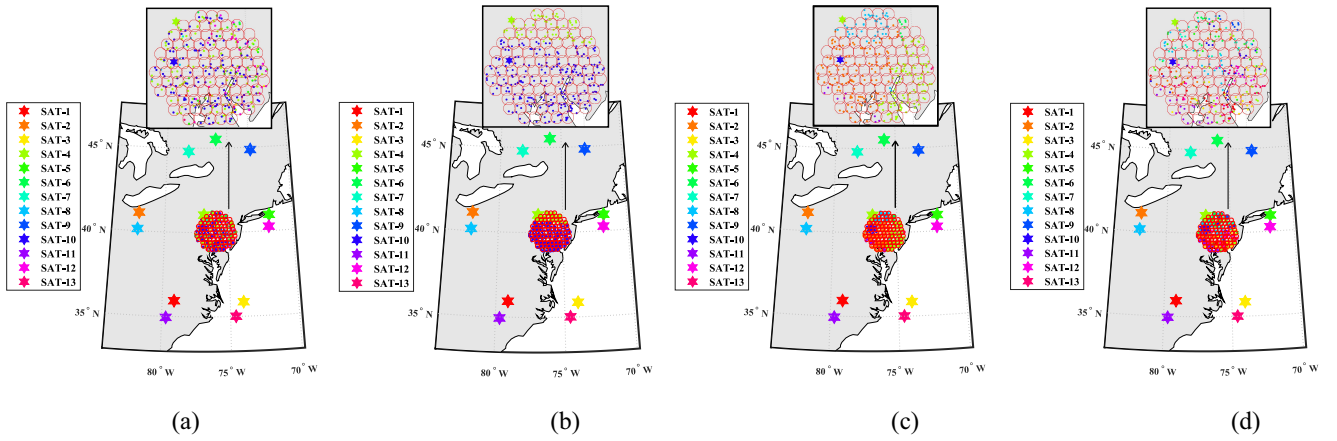


Fig. 5. Illustration of user-satellite association at the first scheduling epoch under (a) *BestChannel*, (b) *MaxElevation*, (c) *MaxTime*, and (d) *Proposed* schemes when 300 UEs are evenly distributed in the area.

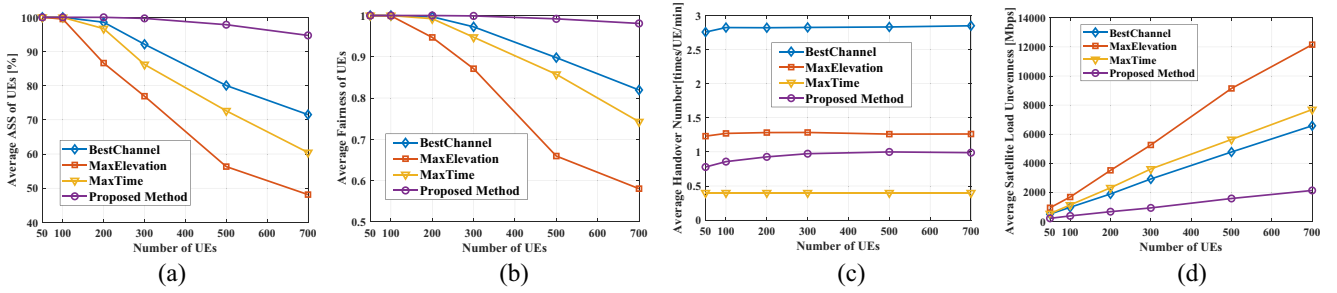


Fig. 6. Performance of (a) average ASS, (b) average fairness, (c) average handover number, and (d) average satellite load unevenness under different user-satellite association schemes when UEs are evenly distributed in the area.

the satellite networks, which is calculated by $\text{mean}_{s \in \mathcal{S}}(\min_{u \in \mathcal{U}} \{ASS_u^s\})$.

- 4) *Average Handover Number*: This metric is calculated by $\sum_{s \in \mathcal{S}, u \in \mathcal{U}} \Delta_u^s / (UST_{\text{epoch}}/60)$, which represents the average number of intersatellite handover per user per minute over the entire simulation duration.
- 5) *Average Satellite Load Unevenness*: This metric is calculated by $\text{mean}_{s \in \mathcal{S}}(\max_{n \in \mathcal{N}_v^s} \{Y_n^s\} - \min_{n \in \mathcal{N}_v^s} \{Y_n^s\})$, where $\mathcal{N}_v^s$ denotes the set of visible satellites by the service area in epoch s and Y_n^s is the traffic load of satellite n . This metric represents the uneven distribution of traffic load among different satellites.

It should be noted that the last two metrics are only related to the user-satellite association scheme.

1) *Performance Under Different UE-Satellite Association Schemes*: We first evaluate the performance of the proposed UE-satellite association approach and other benchmarks, considering two types of user distributions. Meanwhile, the BH pattern design scheme and the RB allocation method both utilize the proposed algorithms (i.e., MDW-BH and SCA method) for fair comparison. Fig. 5 shows the user-satellite association result at the first scheduling epoch with 300 uniformly distributed UEs. It can be observed that there are 13 satellites in the FoV at this moment. UEs are densely accessing 3–4 satellites in *MaxElevation* and *MaxTime*, which can significantly exacerbate the workload of these satellites. In addition, UEs are dispersedly assigned to various satellites in *BestChannel* and the proposed method, balancing the

traffic load among satellites. Fig. 6 shows the performance of different UE-satellite association methods. It can be seen that the average ASS and average fairness of UEs [in Fig. 6(a) and (b)] both equal 1 when the number of UEs is less than 50. This is because the traffic demands are much less than the capacity of satellites, and users are well satisfied regardless of the association strategy adopted. However, as the number of UEs increases, the advantages of the proposed method become evident. In particular, the average ASS and average fairness are more than 0.9, which has a performance gain of 35%–90% compared with other benchmarks. This is because our proposed method considers the load balancing and interference avoidance in Algorithm 1. As Fig. 6(c) depicted, *MaxTime* has the lowest number of handovers around 0.4 handover per UE per minute, since UEs always connect to the satellite with the longest remaining service time. Our proposed method has approximately 1 handover per UE per minute, less than *BestChannel* and *MaxElevation*. This is because Algorithm 1 tries to retain the last UE-satellite connection result from the set of candidate satellites during each scheduling, which can reduce the capacity loss caused by handovers. Besides, as shown in Fig. 6(d), the proposed method outperforms others in terms of average satellite load unevenness, which is reduced by 67%–83%.

Figs. 7 and 8 show the performance under real population distribution. We can observe from Fig. 8(a) and (b) that due to the concentrated user distribution, *MaxElevation* and *MaxTime* cannot balance the traffic demands, degrading the

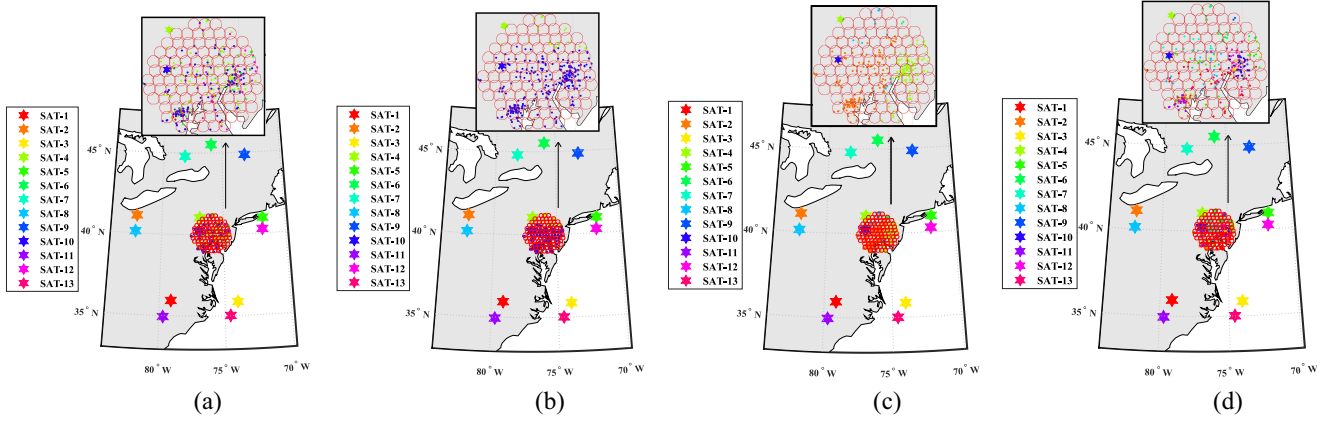


Fig. 7. Illustration of user-satellite association at the first scheduling epoch under (a) *BestChannel*, (b) *MaxElevation*, (c) *MaxTime*, and (d) *Proposed* schemes when 300 UEs are distributed according to population density in the area.

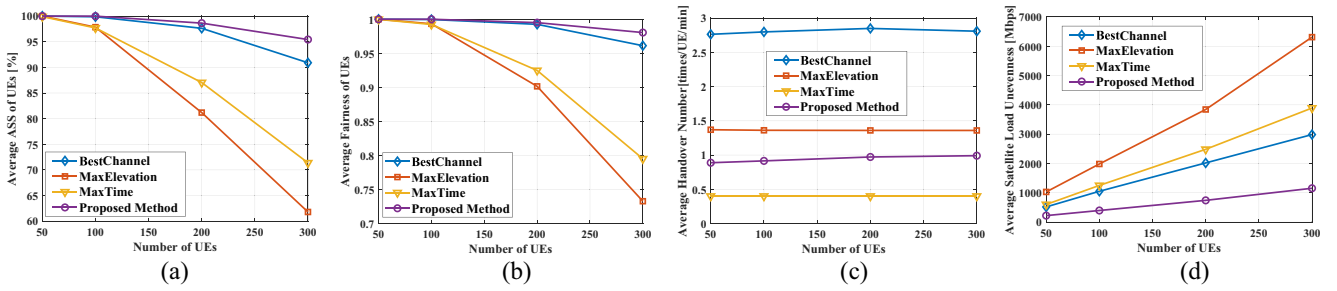


Fig. 8. Performance of (a) average ASS, (b) average fairness, (c) average handover number, and (d) average satellite load unevenness under different user-satellite association schemes when UEs are distributed according to population density in the area.

performance of average ASS and average fairness. Although the *BestChannel* method outperforms the above two benchmarks, it yields more handovers (i.e., 2.7 handovers per UE per minute) as shown in Fig. 8(c). Besides, our method has improved the average ASS by 5% while reducing the number of handovers by 60%, compared to *BestChannel* method. In Fig. 8(d), the proposed method has the best performance on the average satellite load unevenness with around 60% ~ 80% performance gain, since our method can allocate hotspot area's users to different satellites as shown in Fig. 7(d). It should be noted that the performance under the uneven user distribution degrades compared to the uniform user distribution scene, since the interference becomes more severe when users are concentrated.

2) *Performance Under Different BH Pattern Design Schemes*: In this part, we evaluate the performance with different BH pattern design algorithms, where the user-satellite association method and RB allocation method are set as the proposed algorithms. Fig. 9 shows the average ASS and average fairness performance in the even user distribution scenario. It can be seen from Fig. 9(a) that the proposed MDW-BH method outperforms other baselines. Particularly, when the number of UEs is 700, the average ASS of Random-BH, Round-Robin-BH, MDW-BH, and Greedy-BH are 83.3%, 83.6%, 88.7%, and 94.7% respectively. MDW-BH exhibits a performance improvement of 6%–11% since we improve the cell fairness and mitigate interbeam interference by constructing the weighted interference graph. Although Greedy-BH

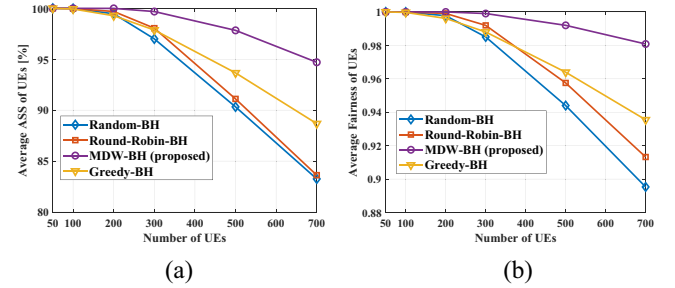


Fig. 9. Performance of (a) average ASS and (b) average fairness under different BH pattern design methods when UEs are evenly distributed in the area.

selects the cells with the lowest service satisfaction to illuminate, the adjacent cells may be simultaneously illuminated, increasing the interference and deteriorating the capacity. In Fig. 9(b), it can be noted that the average fairness of our method equals 0.98, indicating that both users with high or low demands are well satisfied.

Fig. 10 presents the average ASS and average fairness performance in the real population distribution scenario. Despite the users being noneven, our method maintains the best performance regardless of the number of users, which has the 5% performance improvement at least. This is because the interference becomes much complex especially when users are densely distributed. The intrasatellite interference is effectively mitigated in Algorithm 2, where the spotbeams are spatial isolation at the same time.

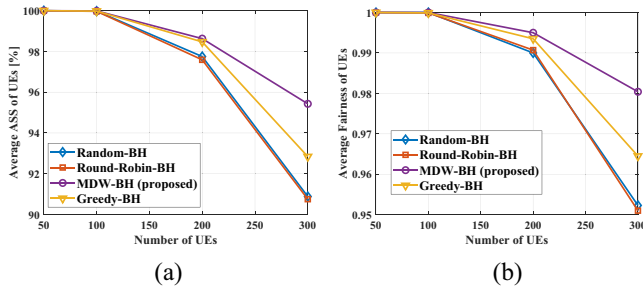


Fig. 10. Performance of (a) average ASS and (b) average fairness under different BH pattern design methods when UEs are distributed according to population density in the area.

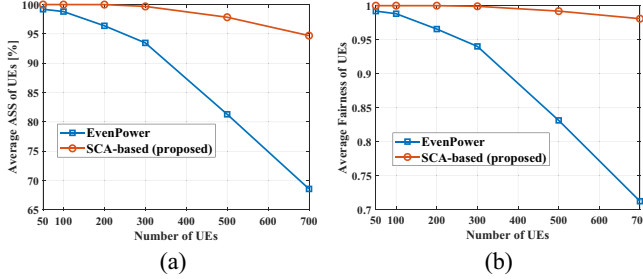


Fig. 11. Performance of (a) average ASS and (b) average fairness under different RB allocation algorithms when UEs are evenly distributed in the area.

3) Performance Under Different RB Allocation Schemes:

In this section, we validate the effectiveness of the proposed SCA-based RB allocation algorithm. The maximum number of SCA iterations is set to 3. Similarly, for a fair comparison, the UE-satellite association method and BH pattern design scheme are fixed as the proposed algorithms. Fig. 11 shows the performance of average ASS and average fairness when UEs are evenly distributed. It can be observed that SCA-based method exhibits superior performance compared with *EvenPower*. Furthermore, as the number of users increases, the performance gain becomes more pronounced. Specifically, the average ASS has been improved by 2%–35% when the number of UEs grows from 100 to 700. Although *EvenPower* allocates RBs to the UEs with lower ASS, the transmit power on each RB is evenly distributed without considering the channel status of each RB, which may decrease the achievable rate of UEs. On the contrary, SCA-based algorithm can adjust iteratively the RB allocation and power allocation according to UEs' demand and channel conditions.

Fig. 12 presents the performance with nonuniform UE distribution. It can be seen that when there are 100 UEs, SCA-based method fully satisfies the traffic requirements of all users, while the average ASS and fairness of *EvenPower* are just 0.95, indicating that 5% UEs are still not satisfied. This phenomenon demonstrates that when UEs are relatively clustered, interference in hotspot areas becomes more severe, posing a challenge to the RB allocation algorithms. Therefore, the transmit power of UEs needs to be accurately controlled.

4) *Performance With Different Number of RBs:* In order to illustrate the impact of different numbers of RBs on service performance, we simulate scenarios with the number of RBs

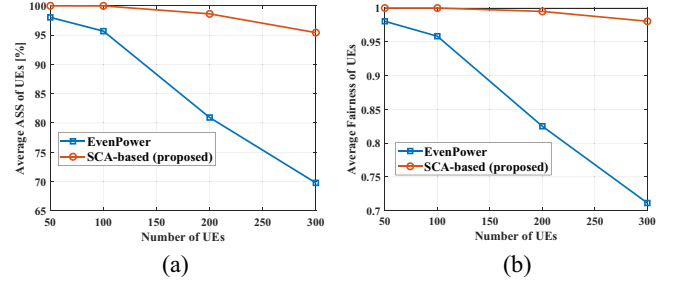


Fig. 12. Performance of (a) average ASS and (b) average fairness under different RB allocation algorithms when UEs are distributed according to population density in the area.

ranging from 1 to 10. Besides, the number of UEs is set to 300, distributed according to population density. Fig. 13 shows the performance of average ASS, average fairness, and the worst UE's service satisfaction with different number of RBs under the proposed method. With the increase in the number of RBs, all metrics show improvements. Specifically, average ASS, average fairness, and the worst UE's service satisfaction improve by approximately 7.8%, 8%, and 40%, respectively. We can observe from Fig. 13(a) and (b) that as the number of RBs exceeds 5, the growth rate of average ASS and average fairness will slow down. This is because with more RBs, more UEs will access the network, resulting in a potential increase in interference between UEs. It can be seen from Fig. 13(c) that when the number of RB equals 1, the worst UE's service satisfaction is 0%. This is because the total number of available RB resources in the entire BHTP is $8 \times 15 = 120$, which is less than the number of UEs (i.e., 300). This inevitably leads to some UEs not being allocated RBs, resulting in them not being effectively served. On the other hand, when the number of RBs is equal to 10, worst UE's service satisfaction can reach 40%, which means that the finer the RB resource granularity, the higher the resource utilization rate and the better the user service quality.

5) Performance With Different Number of Satellites:

Finally, we validate the scalability of the proposed method to very large constellations. The number of satellites ranges from 720 to 21 600 and the number of RBs is set to 5. Fig. 14 shows the curves of four different indicators changing with the number of satellites when 300 UEs are uneven distributed. In Fig. 14(a) and (b), it can be seen that average ASS and average fairness have the same change trend, and when the number of satellites exceeds 5000, the average ASS almost reaches 100% and the average fairness approaches 1, indicating that most UEs' traffic demands are well satisfied. Fig. 14(c) reveals an interesting phenomenon. The growth rate of the worst UE's service satisfaction reaches a turning point when the number of satellites exceeds 5000, which illustrates that although expanding the constellation will provide more service opportunities, it also leads to heavy intersatellite interference, weakening the service capabilities of satellite network. In addition, we can observe from Fig. 14(d) that as the size of the constellation increases, UEs have more access options, resulting in an increase in the number of intersatellite handovers. Therefore, satellite operators need to rationally

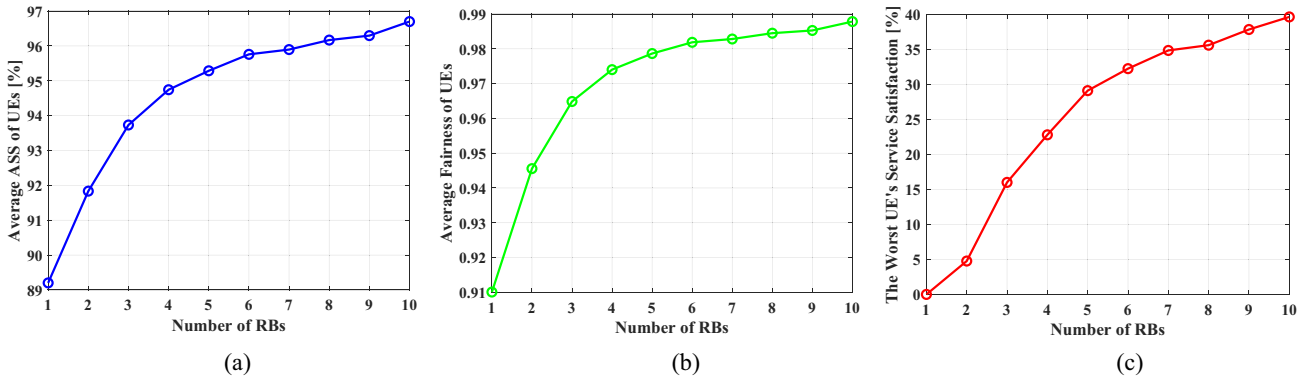


Fig. 13. Performance of (a) average ASS, (b) average fairness, and (c) worst UE's service satisfaction with different number of RBs under the proposed method when 300 UEs are distributed according to population density in the area.

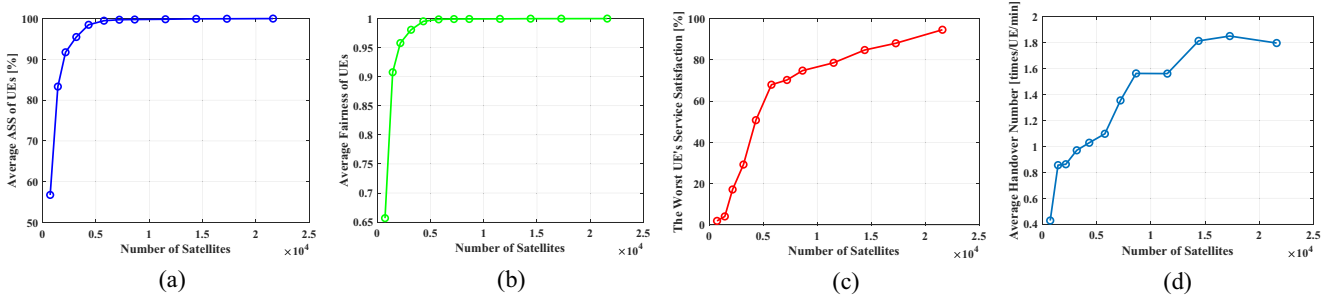


Fig. 14. Performance of (a) average ASS, (b) average fairness, (c) worst UE's service satisfaction, and (d) average handover number with different number of satellites under the proposed method when 300 UEs are distributed according to population density in the area.

plan the number of constellations to adapt to traffic demands. Blindly increasing the scale of constellations may not be worth the gain.

C. Complexity Analysis

Without loss of generality, we analyze the computational complexity of one scheduling epoch as follows.

- 1) *The Complexity of User-Satellite Association*: The user-satellite association is solved in Algorithm 1 every epoch. The complexity of Algorithm 1 mainly relies on the iteratively searching for a candidate satellite set of each UE, which can be represented as $\mathcal{O}(U^2 N_{\max} \lfloor \vartheta_{TH}/5 \rfloor)$ in the worst case. $N_{\max}$ represents the maximum number of visible satellites by UEs with $N_{\max} \ll N$.
- 2) *The Complexity of BH Pattern Design*: At each timeslot, each visible satellite will execute Algorithm 2 to obtain the BH pattern. The complexity of Algorithm 2 depends on the iteratively searching for the vertex with maximum weight slot-by-slot, where the computational complexity for each timeslot can be expressed as $\mathcal{O}(MK)$ in the worst case. Hence, the computational complexity of BH pattern design during one scheduling epoch can be represented as $\mathcal{O}(N_{\max} TMK)$.
- 3) *The Complexity of RB Allocation*: At each timeslot, each visible satellite will execute Algorithm 3 to obtain the RB allocation result. The complexity of Algorithm 3 hinges on the interior point method used in SCA process [48]. Since the convex approximate

program $\mathcal{P}_4$ has a total of $2UKF$ scalar decision variables and the maximum iterations is $I_{\max}$, the worst case computational complexity of SCA for each satellite is $\mathcal{O}(I_{\max}(2UKF)^3)$. Therefore, the complexity of RB allocation is estimated as $\mathcal{O}(N_{\max} I_{\max}(2UKF)^3)$.

As a result, the total complexity can be obtained as $\mathcal{O}(N_{\max}(U^2 \lfloor \vartheta_{TH}/5 \rfloor + TMK + I_{\max}(2UKF)^3))$, which is linear with the maximum number of visible satellites $N_{\max}$, indicating that the proposed method has good scalability for mega-constellations.

V. CONCLUSION

This article proposes a universal return-link BH scheduling framework for NGSO mega-constellation communication systems with time-varying topology and uneven traffic. We formulate the dynamic multisatellite BH problem to the periodically optimize user-satellite association, BH pattern, and RB allocation. First, we propose a low-complexity heuristic user-satellite association strategy to balance the traffic load among satellites while reducing the intersatellite handover. Second, the weighted interference graph is constructed to assist the BH design slot by slot, where a greedy algorithm based on maximum weight search is proposed. Third, in order to make the RB allocation problem tractable, we present an SCA-based iterative algorithm to refine the RB and power allocation results in the polynomial-time complexity. Extensive simulations have proven the superiority of the proposed method on improving the average service satisfaction (ASS) of users

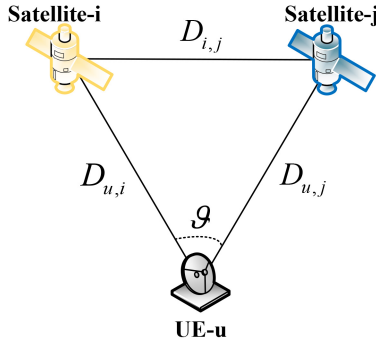


Fig. 15. Illustration of the Appendix.

and reducing the handover overhead in the dynamic satellite network.

APPENDIX

PROOF OF PROPOSITION 1

Proof: As shown in Fig. 15, given arbitrary one UE u and two visible satellites i and j , denote $D_{u,i}$ and $D_{u,j}$ by the distance from UE u to satellites i and j , respectively. According to the cosine theorem, we can obtain $\cos(\vartheta) = (D_{u,i}^2 + D_{u,j}^2 - D_{i,j}^2) / (2D_{u,i}D_{u,j})$, where $r_{\min} \leq D_{u,i} \leq r_{\max}$ and $r_{\min} \leq D_{u,j} \leq r_{\max}$. Furthermore, the difference between $\cos(\vartheta)$ and $\cos(\vartheta_{TH})$ can be expressed as

$$\begin{aligned} \cos(\vartheta) - \cos(\vartheta_{TH}) &= \frac{D_{u,i}^2 + D_{u,j}^2 - D_{i,j}^2}{2D_{u,i}D_{u,j}} - \cos(\vartheta_{TH}) \\ &= \frac{D_{u,i}^2 + D_{u,j}^2 - D_{i,j}^2 - 2D_{u,i}D_{u,j}\cos(\vartheta_{TH})}{2D_{u,i}D_{u,j}}. \end{aligned} \quad (19)$$

Denote $f(D_{u,i}, D_{u,j}) = D_{u,i}^2 + D_{u,j}^2 - 2D_{u,i}D_{u,j}\cos(\vartheta_{TH})$. It can be easily seen that $f(D_{u,i}, D_{u,j})$ is a convex function due to $\nabla^2 f(D_{u,i}, D_{u,j}) \geq 0$. Thus, the maximum value of $f(D_{u,i}, D_{u,j})$ is obtained on the boundary, which can be represented as $f_{\max} = \max\{f(r_{\max}, r_{\min}), f(r_{\min}, r_{\max})\}$. Substituting this in the right-hand side of (19) yields

$$\begin{aligned} \cos(\vartheta) - \cos(\vartheta_{TH}) &= \frac{f(D_{u,i}, D_{u,j}) - D_{i,j}^2}{2D_{u,i}D_{u,j}} \\ &\leq \frac{f_{\max} - D_{i,j}^2}{2D_{u,i}D_{u,j}}. \end{aligned} \quad (20)$$

Therefore, if $D_{i,j} > \sqrt{f_{\max}} = D_{TH}$, we can obtain $\cos(\vartheta) - \cos(\vartheta_{TH}) < 0$. This implies that $\vartheta > \vartheta_{TH}$ and completes the proof. ■

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